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Inventor(s) HUFNAGEL; Martin et al.

REDUCTION OF MONOMERIC ISOCYANATES BY POROUS MATERIAL

Abstract

Disclosed herein is a process for the reduction of the content of residual monomeric isocyanate in a polyurethane prepolymer containing the residual monomeric isocyanate including the step of contacting the polyurethane prepolymer with a porous material having pores formed by a framework containing isocyanate reactive functional groups under conditions allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores to covalently bind the reacted isocyanate, where the porous material is a porous metal-organic framework material including at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to the metal ion. Also disclosed herein is a prepolymer composition including the porous material and a method of using the porous material for removal of residual monomeric isocyanate in the prepolymer.

Inventors: **HUFNAGEL; Martin (Lemfoerde, DE), LINNENBRINK; Martin (Lemfoerde, DE), ALBUERNE; Julio (Lemfoerde, DE), MATHIEU; Thomas (Lemfoerde, DE), GORDILLO BOLONIO; Alvaro (Barcelona, ES), LIESE; Julia (Lemfoerde, DE), MARX; Stefan (Ludwigshafen, DE), Clade; Stefanie (Ludwigshafen, DE)**

Applicant: **BASF SE** (Ludwigshafen, DE)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Phase Application of International Patent Application No. PCT/EP2022/052865, filed Feb. 7, 2022, which claims priority to EP patent application No. 21156620.3, filed Feb. 11, 2021, each of which is hereby incorporated by reference herein.

[0002] The present invention relates to a process for the reduction of the content of (removal of) residual monomeric isocyanate in a polyurethane prepolymer containing said residual monomeric isocyanate comprising the step of contacting the polyurethane prepolymer with a porous material having pores formed by a framework containing isocyanate reactive functional groups under conditions allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores in order to covalently bind the reacted isocyanate. The invention also relates to a prepolymer composition comprising said porous material and the use of said porous material for removal residual monomeric isocyanate in the prepolymer.
[0003] Polyurethane prepolymers based on the reaction of polyisocyanates and polyols typically possess a given amount of monomeric isocyanates. These prepolymers in general are used for manifold applications e.g. foamed and compact elastomers, sealants, coatings and adhesives in 1-component and 2-component formulations.
[0004] The hazardous nature of monomeric isocyanates leads to a severe increase of regulatory restrictions from governmental institutions. In 2018 the ECHA suggested to tighten the restrictions of use for diisocyanates significantly. E.g. for industrial use, the handling of products with residual monomeric isocyanates will require extensive safety trainings and additional technical measures to keep isocyanate levels in air low. Further, any private use of isocyanate based products in do-it-yourself applications like sealants or can foams is already to date restricted and further limitations are expected. To minimize the hazardous potential of these products, a limit of 0.1 wt.-% of monomeric isocyanate was defined (“monomer free”). Products with contents below 0.1 wt.-% of aromatic or aliphatic isocyanate enable easy and safe handling of the monomer-free products both for industrial and private users.
[0005] One approach to obtain low monomeric isocyanate content is known in the art by designing an appropriate chemical protocol for the synthesis of the polyurethane prepolymer.
[0006] EP 2 155 797 B1 describes a process for the production of isocyanate prepolymers based on toluene diisocyanate having a content of free monomeric toluene diisocyanate content of not more than 0.1 wt.-% using polyether polyols, wherein a certain index based on the hydroxyl number of the polyether is not exceeded.
[0007] EP 951 493 B1 describes a low-monomer polyurethane prepolymer containing free isocyanate groups obtained by reacting one or more polyhydric alcohols and two or more diisocyanates, wherein a ratio of a total number of first and second isocyanate groups of non-

symmetrical diisocyanate to a number of isocyanate groups of a second diisocyanate is greater than 6:1.
[0008] EP 150 444 B1 describes a process for the production of isocyanate-terminated polyurethane prepolymers based on diisocyanates of different reactivity, wherein in a first reaction step carried out in the absence of other diisocyanates, 2,4-tolylene diisocyanate is reacted and in a second reaction step, a symmetrical, dicyclic diisocyanate which is more reactive by comparison with the less reactive isocyanate groups of the 2,4-tolylene diisocyanate used in the first reaction step is added.
[0009] However this approach has the drawbacks that it contains many steps causing high costs and a very high level of reaction control is required.
[0010] A further approach to reduce monomeric isocyanate content is the distillation of the prepolymer.
[0011] WO 03/046 040 A1 describes a process for the preparation of prepolymers comprising isocyanate and urethane groups, comprising the step of removing monomeric diisocyanates by distilling the reaction product from step a) by means of a short-path evaporator.
[0012] EP 1 241 197 A1 describes a method for forming an isocyanate-functional prepolymer having a low residual isocyanate content by passing the reaction mixture comprising prepolymer and unreacted isocyanate through a short-path evaporator to remove unreacted isocyanate to a level of less than 0.15% by weight.
[0013] DE 199 39 840 A1 describes a process for producing prepolymers which contain isocyanate terminal groups, wherein the reaction mixture is subsequently subjected to distillation, optionally with the addition of an entraining agent.
[0014] However, this approach has the drawback that an expensive distillation setup is required and that it can only be used for prepolymers that are stable under the distillation conditions and show low viscosities.
[0015] A further approach uses adsorbent material in order to remove residual monomeric isocyanate.
[0016] U.S. Pat. No. 4,061,662 describes a method of reducing residual unreacted tolylene diisocyanate content in a prepolymer reaction product of toluene diisocyanate with a polyoxyalkylene polyol comprising allowing said prepolymer to flow through a column packed with absorbent type X zeolite molecular sieves. However, the additional separation step results in an increased operating expense as described above for the distillation approach and the effectiveness is limited due to the high amount of zeolite that is needed. A further drawback of this and the distillation approach is that the amount of material is reduced due to the removal of the monomeric isocyanates.
[0017] EP 2 439 220 A1 describes a carrier material which has functional groups on its surface which are reactive towards isocyanate groups, for reducing the proportion of isocyanate-containing monomers in compositions containing at least one isocyanate-containing polyurethane polymer and at least one isocyanate-containing monomer. This is accomplished by surface derivatization of amorphous silica by amino groups. However surface modification results in a limited active surface compared to the silica particles and is non-selective as not only residual monomeric isocyanates can react with the amino functions on the surface but also other amino-reactive components of the polymer, including terminal isocyanate functional groups of the polymer itself as well as other oligo- and polymeric isocyanates. The huge drawback of this approach is the tremendous increase in viscosity after contacting theses functionalized silica with isocyanate containing prepolymers.
[0018] International patent application with application N° PCT/EP2020/079866 describes the removal of residual monomeric aliphatic isocyanate from a reaction mixture comprising aliphatic polyisocyanate using a scavenger comprising a zeolitic material.
[0019] Thus, an object of the present invention is to provide a method and polymer composition overcoming the above disadvantages.

[0020] The object is achieved by a process for the reduction of the content of (removal of) residual monomeric isocyanate in a polyurethane prepolymer containing said residual monomeric isocyanate comprising the step of [0021] (i) contacting the polyurethane prepolymer with a porous material having pores formed by a framework containing isocyanate reactive functional groups under conditions allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores in order to covalently bind the reacted isocyanate, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.

[0022] The object is also achieved by a polyurethane prepolymer composition, the composition comprising a polyurethane prepolymer comprising a porous material having pores formed by a framework containing isocyanate reactive functional groups, wherein the reactive functional groups are reacted with monomeric isocyanate in at least a fraction of the pores, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.

[0023] The object is also achieved by the use of a porous material having pores formed by a framework containing isocyanate reactive functional groups to remove residual monomeric isocyanate in a polyurethane prepolymer by chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.

[0024] Surprisingly, it was found that the use of a porous metal-organic framework (MOF) material having pores formed by a framework containing isocyanate reactive functional groups under conditions allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores in order to covalently bind the reacted isocyanate, especially Basolite A 120 results in a substantially improved ability to capture the free monomer from TDI and MDI prepolymers. The functional group within the pores of the porous material allows a selective reaction with monomeric isocyanate compared to oligomeric or polymeric isocyanate comprising substances. To remove the monomeric isocyanates, the prepolymer can-by way of example-be simply mixed with the porous material, annealed at 60° C. for 1-2 hours and kept at room temperature for 24 hours. The combination of the porous material that has an intrinsic size selectivity towards polymeric and monomeric components together with reactive groups inside the pores turned out to be the most efficient absorbent for monomer removal in isocyanate prepolymers. Further benefit is the moderate viscosity increase of the prepolymers containing the porous material, especially Basolite A 120, which is significantly lower than for the amino-functional fumed silicas.

[0025] Polyurethane prepolymers prepared from monomeric isocyanate are known in the art. In view of the selectivity of the porous material isocyanate groups containing polyurethane prepolymers are suitable.

[0026] The polyurethane prepolymers can be prepared by reaction of polyisocyanates with compounds having at least two isocyanate-reactive hydrogen atoms.

[0027] Suitable polyisocyanates include, for example, aliphatic, cycloaliphatic, aromatic and particularly aromatic diisocyanates. The following are specifically named by way of example: aliphatic diisocyanates, such as hexamethylene 1,6-diisocyanate, 2-methylpentamethylene 1,5-diisocyanate, 2-ethylbutylene 1,4-diisocyanate or mixtures of at least two of said C.sub.6-alkylene diisocyanates, pentamethylene 1,5-diisocyanate and butylene 1,4-diisocyanate, cycloaliphatic diisocyanates, such as 1-isocyanato-3,3,5-trimethyl-5-isocyanatomethylcyclohexane (isophorone diisocyanate), 1,4-cyclohexane diisocyanate, 1-methyl-2,4- and -2,6-cyclohexane diisocyanate, as well as the corresponding isomer mixtures, 4,4′-, 2,4′- and 2,2′-dicyclo-hexylmethane diisocyanate, as well as the corresponding isomer mixtures and preferably aromatic diisocyanates, such as 1,5-

naphthylene diisocyanate (1,5-NDI), 2,4- and 2,6-tolylene diisocyanate (TDI) as well as their mixtures, 2,4′-, 2,2′-, and preferably 4,4′-diphenylmethane diisocyanate (MDI) as well as mixtures of at least two of these isomers, having two or more aromatic systems. Suitable isocyanate components can also include derivatives of said isocyanates, such as uretdione, urea, biuret or isocyanurate and mixtures thereof.

[0028] According to the process of the present invention said polyisocyanates represent residual monomeric isocyanate to be removed in the polyurethane prepolymer. The term “removal” means according to the present invention any reduction of the content of residual monomeric isocyanate resulting from the contacting with the porous material. Preferably, the content of residual isocyanate in the prepolymer according to the present invention is at most 0.1 wt.-% (percent by weight) based on the total weight of the prepolymer. The content of residual monomeric isocyanate is preferably determined by using high performance liquid chromatography (HPLC) with UV detection of samples derivatized with 1-(2-Piridyl) piperazine.

[0029] In general polyisocyanates used in the preparation of the polyurethane prepolymer may have a molecular weight of less than 500 g/mol, preferably less than 400 g/mol, even more preferably less than 300 g/mol. The prefix “poly” in the term polyisocyanate refers to the fact that the isocyanate has at least two isocyanate groups and does not refer to a polymeric nature in weight. The polyisocyanates represent monomers and thus are not oligomers or polymers.

[0030] Accordingly, the residual monomeric isocyanate is preferably an aromatic or aliphatic isocyanate. Thus, in one embodiment of the present invention the residual monomeric isocyanate is an aromatic isocyanate. In another embodiment of the invention the residual monomer is an aliphatic isocyanate. In an additional embodiment of the invention the residual monomer is a mixture of aliphatic and aromatic monomeric isocyanates.

[0031] A preferred residual monomeric isocyanate is MDI in isomeric pure form or as mixture of 2,2′-MDI, 4,4′-MDI and 2,4′-MDI. Another preferred residual monomeric isocyanate is TDI in isomeric pure form or as mixture of 2,4-TDI and 2,6-TDI.

[0032] Suitable compounds having at least two isocyanate-reactive hydrogen atoms preferably have at least two hydroxyl and/or amino groups in the molecule. Particularly suitable compounds have a molecular weight Mn in the range from 60 to 10 000 g/mol. The compounds having at least two isocyanate-reactive hydrogen atoms are particularly preferably selected from the group consisting of polyfunctional alcohols, polyetheralcohols, polyesteralcohols, polyetherpolyamines, hydroxyl-containing polycarbonates, hydroxyl-containing polyacetals and any mixtures of at least two thereof. Polyfunctional alcohols and polyether-alcohols, as well as mixtures thereof, are particularly preferred.

[0033] Examples of suitable polyfunctional alcohols are alkanediols having from 2 to 10, preferably from 2 to 6 carbon atoms, as well as higher alcohols, such as glycerol, trimethylolpropane or pentaerythritol. Furthermore, natural polyols can be used, such as castor oil.

[0034] The polyetheralcohols are preferably from di- to octafunctional. Their preparation is usually carried out by addition of alkylene oxides, in particular ethylene oxide and/or propylene oxide, to H-functional starters. The alkylene oxides can be used individually, in sequence or as a mixture. Possible starters include, for example, water, diols, triols, alcohols of higher functionality, sugar alcohols, aliphatic or aromatic amines or aminoalcohols.

[0035] Polyetheralcohols having an average molecular weight in the range from 500 to 10 000 g/mol and an average OH-functionality in the range from 2 to 6 are particularly suitable. Particularly preferred starters for the preparation of these polyetheralcohols are propylene glycol and glycerol. Preferred alkylene oxides are ethylene oxide and propylene oxide.

[0036] Polyesteralcohols having average molecular weights in the range from 1 000 to 10 000 g/mol and an average OH-functionality in the range from 2 to 3 are also preferred. Polyesteralcohols-based on adipic acid are particularly preferred. The preparation of the prepolymers is carried out, as explained, by reaction of the polyisocyanates with the compounds

having at least two isocyanate-reactive hydrogen atoms.
[0037] Suitable catalysts, which in particular speed the reaction between the isocyanate groups of the polyisocyanates and the hydroxyl groups of the polyalcohols, include the known and conventional strongly basic amines and also organic metal compounds such as titanic esters, iron compounds such as iron(III) acetylacetonate, tin compounds, e.g. tin(II) salts of organic carboxylic acids, or the dialkyltin(IV) salts of organic carboxylic acids or mixtures of at least two of the named catalysts, bismuth salts of organic carboxylic acids as well as synergistic combinations of strongly basic amines and organic metal compounds. The catalysts can be used in the usual quantities, for example from 0.002 to 5% by weight, based on the polyalcohols.
[0038] The reaction can be carried out continuously or batchwise in the usual reactors, for example the known tubular or stirred-tank reactors, preferably in the presence of the usual catalysts, which speed the reaction of the OH-functional compounds with isocyanate groups, with or without inert solvents, i.e. compounds that do not react with the isocyanates and OH-functional compounds.
[0039] For instance, a polyurethane prepolymer was prepared by reacting toluene diisocyanate (TDI) with a ratio of 2,4-TDI:2,6-TDI=80:20 with poly(tetramethylene ether glycol) with an OH value of 112 mg KOH/g in the presence of diglycol-bis-chloroformiat (DIBIS). The prepolymer has an NCO content of 6.09%, free monomer content was 0.4% 2,4-TDI and 2.1% 2,6-TDI and a viscosity of 11.4 Pas at 25° C.
[0040] Another exemplary polyurethane prepolymer was prepared by reacting methylene diphenyl diisocyanate (MDI) with a ratio of 4,4′-MDI:2,4′-MDI:2,2′-MDI=1:1:0.05 with poly(propylene glycol) with an OH value of 104 mg KOH/g in the presence of Diglycol-bis-chloroformiat (DIBIS). The prepolymer has an NCO content of 5.27%, free monomer content of 2.1% 4,4′-MDI, 4.7% 2,4′-MDI and 0.4% 2,2′-MDI and a viscosity of 2,8 Pas at 60° C.
[0041] The prepolymers are customarily used in the preparation of polyurethanes. To this end, the prepolymers are reacted with compounds that can react with isocyanate groups. The compounds that can react with isocyanate groups include, for example, water, alcohols, amines or compounds having mercapto groups. The polyurethanes can be used as foams, coatings, adhesives, in particular melt-applied adhesives, paints, as well as compact or cellular elastomers.
[0042] The polyurethane prepolymer is contacted with a porous MOF material. Said porous MOF material has pores formed by a framework. Within the pores the framework contains isocyanate reactive functional groups capable of allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores in order to covalently bind the reacted isocyanate. The covalently binding results in reduction of residual monomeric isocyanate content.
[0043] The porous MOF material has pores, wherein the mean pore size of the porous material before contacting, i.e. before reaction with residual monomeric isocyanate, is preferably in the range from 0.3 to 9.0 nm. More preferably, the range is from 0.5 to 5.0 nm, even more preferably from 1.0 to 2.5 nm, even more from 1.5 to 2.2 nm, according to DIN ISO 9277:2014-01 (Determination of the specific surface area of solids by gas adsorption-BET method).
[0044] Preferably, the porous MOF material has before the contacting a total pore volume in the range from 0.04 to 4.0 cm.sup.3/g. More preferably, the range is from 0.4 to 2.0, even more preferably 1.0 to 1.8 cm.sup.3/g, according to DIN ISO 9277:2014-01 (Determination of the specific surface area of solids by gas adsorption—BET method).
[0045] Preferably, the porous MOF material has before the contacting a specific surface area BET from 10 to 5700 m.sup.2/g, more preferably from 150 to 5700 m.sup.2/g. More preferably, the range is from 300 to 4500 m.sup.2/g, even more preferably, from 400 to 4500 m.sup.2/g, even more preferably from 1000 to 3300 m.sup.2/g. according to DIN ISO 9277:2014-01 (Determination of the specific surface area of solids by gas adsorption—BET method).
[0046] The isocyanate reactive functional groups are preferably amino, hydroxyl, thiol, epoxy or a mixture of two or more of these groups, preferably amino groups, more preferably primary amino

groups.

[0047] Examples for the at least bidentate organic compound functionalized with one or more amino groups are for example aspartic acid, glutamic acid, arginine, histidine, proline, tryptophane, amino-fumaric acid, amino-maleic acid, amino-tartaric acid, amino-muconic acid, amino-hydroxy-1,4-benzenedicarboxylic acid, amino-dihydroxy-1,4-benzenedicarboxylic acid, amino-imidazole, amino-pyrazole, amino-1,4-butanedicarboxylic acid, amino-1,4-butenedicarboxylic acid, amino-1,6-hexanedicarboxylic acid, amino-decanedicarboxylic acid, amino-1,8-heptadecanedicarboxylic acid, amino-1,9-heptadecanedicarboxylic acid, amino-heptadecanedicarboxylic acid, amino-1,2-benzenedicarboxylic acid, amino-1,3-benzenedicarboxylic acid, amino-2,3-pyridinedicarboxylic acid, amino-pyridine-2,3-dicarboxylic acid, amino-1,3-butadiene-1,4-dicarboxylic acid, amino-1,4-benzenedicarboxylic acid, amino-p-benzenedicarboxylic acid, amino-imidazole-2,4-dicarboxylic acid, 4,4′-diaminophenylmethane-3,3′-dicarboxylic acid, amino-pyridine-2,6-dicarboxylic acid, amino-thiophene-3,4-dicarboxylic acid, amino-perylene-3,9-dicarboxylic acid, amino-perylenedicarboxylic acid, amino-3,6-dioxaoctanedicarboxylic acid, amino-3,5-cyclohexadiene-1,2-dicarboxylic acid, amino-octadicarboxylic acid, amino-pentane-3,3-carboxylic acid, amino-1,1′-dinaphthyldicarboxylic acid, amino-phenylindanedicarboxylic acid, amino-1,4-cyclohexanedicarboxylic acid, amino-naphthalene-1,8-dicarboxylic acid, amino-2,-benzoylbenzene-1,3-dicarboxylic acid, amino-1,4-naphthalenedicarboxylic acid, amino-2,6-naphthalenedicarboxylic acid, amino-1,3-adamantanedicarboxylic acid, amino-1,8-naphthalenedicarboxylic acid, amino-2,3-naphthalenedicarboxylic acid, amino-8-methoxy-2,3-naphthalenedicarboxylic acid, amino-8-nitro-2,3-naphthalenecarboxylic acid, amino-8-sulfo-2,3-naphthalenedicarboxylic acid, amino-anthracene-2,3-dicarboxylic acid, amino-imidazole-4,5-dicarboxylic acid, amino-4-cyclohexene-1,2-dicarboxylic acid, amino-tetradecanedicarboxylic acid, amino-1,7-heptadicarboxylic acid, amino-5-hydroxy-1,3-benzenedicarboxylic acid, amino-2,5-dihydroxy-1,4-dicarboxylic acid, amino-pyrazine-2,3-dicarboxylic acid, amino-furan-2,5-dicarboxylic acid, amino-2,5-pyridinedicarboxylic acid, amino-cyclohexene-2,3-dicarboxylic acid, amino-1,3-benzenedicarboxylic acid, amino-2,6-pyridinedicarboxylic acid, amino-anthraquinone-1,5-dicarboxylic acid, amino-3,5-pyrazoledicarboxylic acid, amino-2-nitrobenzene-1,4-dicarboxylic acid, amino-heptane-1,7-dicarboxylic acid, amino-cyclobutane-1,1-dicarboxylic acid, amino-1,14-tetradecanedicarboxylic acid, amino-5,-6-dehydronorbornane-2,3-dicarboxylic acid, amino-5-ethyl-2,3-pyridinedicarboxylic acid or amino-camphordicarboxylic acid.

[0048] For example, the at least bidentate organic compound can be derived from a tricarboxylic acid that already has at least one isocyanate reactive functional group or that can serve as starting material for the preparation of a tricarboxylic acid by derivatization. Examples of tricarboxylic acids are amino-2-hydroxy-1,2,3-propanetricarboxylic acid, amino-1,2,3-, amino-1,2,4-benzenetricarboxylic acid, amino-1,2,4-butanetricarboxylic acid, amino-1,3,5-benzenetricarboxylic acid, amino-1-hydroxy-1,2,3-propanetricarboxylic acid, amino-benzene-trisbenzoic acid.

[0049] Examples of an at least bidentate organic compound derived from a tetracarboxylic acid that can serve as starting material for the preparation of a tetracarboxylic acid by derivatization, are

[0050] amino-1,2,4,5-benzenetetracarboxylic acid, amino-1,2,5,6-hexanetetracarboxylic acid, amino-1,2,7,8-octanetetracarboxylic acid, amino-1,4,5,8-naphthalenetetracarboxylic acid, amino-1,2,9,10-decanetetracarboxylic acid, amino-benzophenonetetracarboxylic acid, amino-3,3′,4,4′-benzophenonetetracarboxylic acid.

[0051] In a preferred embodiment the polyurethane prepolymer is liquid under the conditions allowing the chemical reaction. As mentioned above the viscosity of a prepolymer can be influenced by the polyisocyanate as starting material and residual monomeric isocyanate. Further viscosity influencing factors are known by the practitioner in the art. Within the meaning of the present invention the term “liquid” is preferably used in case the viscosity is in the range from 0.1 to 10.sup.7 mPas. The viscosity can be determined with a Haake-Mars rheometer with plate-plate geometry in rotation and 1 mm gap width at a shear rate of 10 and 40 s.sup.−1 at the given

temperature.
[0052] Preferably, the ratio (w/w) polyurethane prepolymer to porous material is in the range of from 100:0.1 to 100:50, more preferably from 100:0.1 to 100:20, even more preferably from 100:0.1 to 100:10, even more preferably from 100:0.1 to 100:5.
[0053] Preferably, the residual monomer isocyanate content of the polyurethane prepolymer is reduced by at least 50%, more preferably by at least 90%, even more preferably by at least 95%, even more preferably by at least 99% in step (i).
[0054] The isocyanate reactive functional groups contained in the pores of the porous material can be introduced by using a porous material without said groups and subsequently derivatization of the porous material in order to introduce the functional groups. Additionally, the reactive functional groups may be present as a part of one or more framework forming component. The latter is preferred.
[0055] It is preferred that the porous metal-organic framework material comprising the at least one metal ion and the at least one at least bidentate organic compound, which is coordinately bound to said metal ion is prepared by contacting the at least one metal ion with the at least one at least bidentate organic compound, wherein the at least one at least bidentate organic compound already contains the isocyanate reactive functional groups.
[0056] Thus, metal-organic framework material is preferably used, wherein the at least one at least bidentate organic compound already comprises isocyanate reactive functional groups or wherein the at least one at least bidentate organic compound, already known as starting material for the preparation of metal-organic framework material, can be derivatized in order to introduce isocyanate one or more reactive functional groups. By way of example dicarboxylic acids are known to form metal-organic frameworks with metal ion. Such a dicarboxylic acid can be derivatized by introducing an amino group in order to provide an at least bidentate organic compound in order to form a framework with pores having amino groups therein.
[0057] Such metal-organic frameworks (MOFs) are known in the prior art and are described, for example, in U.S. Pat. No. 5,648,508, EP-A-0 790 253, M. O′Keeffe et al., J. Sol. State Chem., 152 (2000), pages 3 to 20, H. Li et al., Nature 402, (1999), page 276, M. Eddaoudi et al., Topics in Catalysis 9, (1999), pages 105 to 111, B. Chen et al., Science 291, (2001), pages 1021 to 1023, DE-A 101 11 230, DE-A 10 2005 053430, WO-A 2007/054581, WO-A 2005/049892 and WO-A 2007/023134.
[0058] The metal-organic frameworks comprise pores, in particular micropores and/or mesopores. Micropores are defined as pores having a diameter of 2 nm or less and mesopores are defined by a diameter in the range from 2 to 50 nm, in each case in accordance with the definition given in Pure & Applied Chem. 57 (1985), 603-619, in particular on page 606. The presence of micropores and/or mesopores can be checked by means of sorption measurements, with these measurements determining the uptake capacity of the MOF for nitrogen at 77 Kelvin in accordance with DIN ISO 9277:2014-01.
[0059] The specific surface area, calculated according to the Langmuir model of an MOF in powder form is preferably more than 100 m.sup.2/g, more preferably above 300 m.sup.2/g, more preferably more than 700 m.sup.2/g, even more preferably more than 800 m.sup.2/g, even more preferably more than 1000 m.sup.2/g and particularly preferably more than 1200 m.sup.2/g.
[0060] Shaped bodies comprising metal-organic frameworks can have a lower active surface area; but preferably more than 150 m.sup.2/g, more preferably more than 300 m.sup.2/g, even more preferably more than 700 m.sup.2/g.
[0061] The metal component in the framework according to the present invention is preferably selected from groups Ia, IIa, IIIa, IVa to VIIIa and Ib to VIb. Particular preference is given to Mg, Ca, Sr, Ba, Sc, Y, Ln, Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W, Mn, Re, Fe, Ro, Os, Co, Rh, Ir, Ni, Pd, Pt, Cu, Ag, Au, Zn, Cd, Hg, Al, Ga, In, TI, Si, Ge, Sn, Pb, As, Sb and Bi, where Ln represents the lanthanides.

[0062] Lanthanides are La, Ce, Pr, Nd, Pm, Sm, En, Gd, Tb, Dy, Ho, Er, Tm, Yb.

[0063] With regard to the ions of these elements, particular mention may be made of Mg.sup.2+, Ca.sup.2+, Sr.sup.2+, Ba.sup.2+, Sc.sup.3+, Y.sup.3+, Ln.sup.3+, Ti.sup.4+, Zr.sup.4+, Hf.sup.4+, V.sup.4+, V.sup.3+, V.sup.2+, Nb.sup.3+, Ta.sup.3+, Cr.sup.3+, Mo.sup.3+, W.sup.3+, Mn.sup.3+, Mn.sup.2+, Re.sup.3+, Re.sup.2+, Fe.sup.3+, Fe.sup.2+, Ru.sup.3+, Ru.sup.2+, Os.sup.3+, Os.sup.2+, Co.sup.3+, Co.sup.2+, Rh.sup.2+, Rh.sup.+, Ir.sup.2+, Ir.sup.+, Ni.sup.2+, Ni.sup.+, Pd.sup.2+, Pd.sup.+, Pt.sup.2+, Pt.sup.+, Cu.sup.2+, Cu.sup.+, Ag.sup.+, Au.sup.+, Zn.sup.2+, Cd.sup.2+, Hg.sup.2+, Al.sup.3+, Ga.sup.3+, In.sup.3+, Tl.sup.3+, Si.sup.4+, Si.sup.2+, Ge.sup.4+, Ge.sup.2+, Sn.sup.4+, Sn.sup.2+, Pb.sup.4+, Pb.sup.2+, As.sup.5+, As.sup.3+, As.sup.+, Sb.sup.5+, Sb.sup.3+, Sb.sup.+, Bi.sup.5+, Bi.sup.3+ and Bi.sup.+.

[0064] Preferred metals are Mg, Ca, Sr, Ba, Sc, Y, Ln, Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W, Mn, Re, Fe, Ro, Os, Co, Rh, Ir, Ni, Pd, Pt, Cu, Ag, Au, Zn, Cd, Hg, Al, Ga, In, Tl, Si, Ge, Sn, Pb, As, Sb, Bi, and mixtures thereof ND wherein Ln represents a lanthanide, preferably Mg, Al, Y, Sc, Zr, Ti, V, Cr, Mo, Fe, Co, Cu, Ni, Zn, Ln, even more preferably Al, Mg, Fe, Cu and Zn, even more preferably Mg and Al, even more preferably Al.

[0065] The term “at least bidentate organic compound” refers to an organic compound which comprises at least one functional group which is able to form at least two coordinate bonds to a given metal ion and/or a coordinate bond to each of two or more, preferably two, metal atoms.

[0066] As functional groups via which the coordinate bonds mentioned can be formed, particular mention may be made of, for example, the following functional groups —CO.sub.2H, —CS.sub.2H, —NO.sub.2, —B(OH).sub.2, —SO.sub.3H, —Si(OH).sub.3, —Ge(OH).sub.3, —Sn(OH).sub.3, —Si(SH).sub.4, —Ge(SH).sub.4, —Sn(SH).sub.3, —PO.sub.3H, —AsO.sub.3H, —AsO.sub.4H, —P(SH).sub.3, —As(SH).sub.3, —CH(RSH).sub.2, —C(RSH).sub.3, —CH(RNH.sub.2).sub.2, —C(RNH.sub.2).sub.3, —CH(ROH).sub.2, —CH(RCN).sub.2, —C(RCH).sub.3, where R is, for example, preferably an alkylene group having 1, 2, 3, 4 or 5 carbon atoms, for example a methylene, ethylene, n-propylene, i-propylene, n-butylene, i-butylene, tert-butylene or n-pentylene group, or an aryl group comprising 1 or 2 aromatic rings, for example 2 C.sub.6 rings, which may, if appropriate, be fused and may be independently substituted by at least one substituent in each case and/or may comprise, independently of one another, at least one heteroatom such as N, O and/or S. In likewise preferred embodiments, functional groups in which the abovementioned radical R is not present are possible. Such groups are, inter alia, CH(SH).sub.2, —C(SH).sub.3, —CH(NH.sub.2).sub.2, —C(NH.sub.2).sub.3, —CH(OH).sub.2, —C(OH).sub.3, —CH(CN).sub.2 or C(CN).sub.3.

[0067] However, the functional groups can also be heteroatoms of a heterocycle. Particular mention may here be made of nitrogen atoms.

[0068] The at least two functional groups can in principle be any suitable organic compound, as long as it is ensured that the organic compound in which these functional groups are present is capable of forming the coordinate bond and of producing the framework.

[0069] The organic compounds which comprise the at least two functional groups are preferably derived from a saturated or unsaturated aliphatic compound or an aromatic compound or a both aliphatic and aromatic compound.

[0070] The aliphatic compound or the aliphatic part of the both aliphatic and aromatic compound can be linear and/or branched and/or cyclic, with a plurality of rings per compound also being possible. More preferably, the aliphatic compound or the aliphatic part of the both aliphatic and aromatic compound comprises from 1 to 15, more preferably from 1 to 14, more preferably from 1 to 13, more preferably from 1 to 12, more preferably from 1 to 11 and particularly preferably from 1 to 10, carbon atoms, for example 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 carbon atoms. Particular preference is here given to, inter alia, methane, adamantane, acetylene, ethylene or butadiene.

[0071] The aromatic compound or the aromatic part of the both aromatic and aliphatic compound can have one or more rings, for example two, three, four or five rings, with the rings being able to be separate from one another and/or at least two rings being able to be present in fused form. The aromatic compound or the aromatic part of the both aliphatic and aromatic compound particularly preferably has one, two or three rings, with one or two rings being particularly preferred. Furthermore, each ring of the specified compound can independently comprise at least one heteroatom such as N, O, S, B, P, Si, Al, preferably N, O and/or S. The aromatic compound or the aromatic part of the both aromatic and aliphatic compound more preferably comprises one or two C.sub.6 rings which are present either separately or in fused form. Particular mention may be made of benzene, naphthalene and/or biphenyl and/or bipyridyl and/or pyridyl as aromatic compounds.

[0072] The at least bidentate organic compound is more preferably an aliphatic or aromatic, acyclic or cyclic hydrocarbon which has from 1 to 18, preferably from 1 to 10 and in particular 6, carbon atoms and additionally has exclusively 2, 3 or 4 carboxyl groups as functional groups.

[0073] For example, the at least bidentate organic compound is derived from a dicarboxylic acid that already has at least one isocyanate reactive functional group or that can serve as starting material for the preparation of a dicarboxylic acid by derivatization. Examples of dicarboxylic acids are oxalic acid, succinic acid, tartaric acid, 1,4-butanedicarboxylic acid, 1,4-butenedicarboxylic acid, 4-oxopyran-2,6-dicarboxylic acid, 1,6-hexanedicarboxylic acid, decanedicarboxylic acid, 1,8-heptadecanedicarboxylic acid, 1,9-heptadecanedicarboxylic acid, heptadecanedicarboxylic acid, acetylenedicarboxylic acid, 1,2-benzenedicarboxylic acid, 1,3-benzenedicarboxylic acid, 2,3-pyridinedicarboxylic acid, pyridine-2,3-dicarboxylic acid, 1,3-butadiene-1,4-dicarboxylic acid, 1,4-benzenedicarboxylic acid, p-benzenedicarboxylic acid, imidazole-2,4-dicarboxylic acid, 2-methylquinoline-3,4-dicarboxylic acid, quinoline-2,4-dicarboxylic acid, quinoxaline-2,3-dicarboxylic acid, 6-chloroquinoxaline-2,3-dicarboxylic acid, 4,4′-diaminophenylmethane-3,3′-dicarboxylic acid, quinoline-3,4-dicarboxylic acid, 7-chloro-4-hydroxyquinoline-2,8-dicarboxylic acid, diimidecarboxylic acid, pyridine-2,6-dicarboxylic acid, 2-methylimidazole-4,5-dicarboxylic acid, thiophene-3,4-dicarboxylic acid, 2-isopropylimidazole-4,5-dicarboxylic acid, tetrahydropyrane-4,4-dicarboxylic acid, perylene-3,9-dicarboxylic acid, perylenedicarboxylic acid, Pluriol E 200-dicarboxylic acid, 3,6-dioxaoctanedicarboxylic acid, 3,5-cyclohexadiene-1,2-dicarboxylic acid, octadicarboxylic acid, pentane-3,3-carboxylic acid, 4,4′-diamino-1,1′-diphenyl-3,3′-dicarboxylic acid, 4,4′-diaminodiphenyl-3,3′-dicarboxylic acid, benzidine-3,3′-dicarboxylic acid, 1,4-bis(phenylamino)benzene-2,5-dicarboxylic acid, 1,1′-dinaphthyldicarboxylic acid, 7-chloro-8-methylquinoline-2,3-dicarboxylic acid, 1-anilinoanthraquinone-2,4′-dicarboxylic acid, polytetrahydrofuran 250-dicarboxylic acid, 1,4-bis(carboxymethyl)piperazine-2,3-dicarboxylic acid, 7-chloroquinoline-3,8-dicarboxylic acid, 1-(4-carboxy)phenyl-3-(4-chloro)phenylpyrazoline-4,5-dicarboxylic acid, 1,4,5,6,7,7-hexachloro-5-norbornene-2,3-dicarboxylic acid, phenylindanedicarboxylic acid, 1,3-dibenzyl-2-oxoimidazolidine-4,5-dicarboxylic acid, 1,4-cyclohexanedicarboxylic acid, naphthalene-1,8-dicarboxylic acid, 2,-benzoylbenzene-1,3-dicarboxylic acid, 1,3-dibenzyl-2-oxoimidazolidine-4,5-cis-dicarboxylic acid, 2,2′-biquinoline-4,4′-dicarboxylic acid, pyridine-3,4-dicarboxylic acid, 3,6,9-trioxaundecanedicarboxylic acid, hydroxy-benophenonedicarboxylic acid, Pluriol E 300-dicarboxylic acid, Pluriol E 400-dicarboxylic acid, Pluriol E 600-dicarboxylic acid, pyrazole-3,4-dicarboxylic acid, 2,3-pyrazinedicarboxylic acid, 5,6-dimethyl-2,3-pyrazinedicarboxylic acid, 4,4′-diamino (diphenyl ether) diimide-dicarboxylic acid, 4,4′-diaminodiphenylmethanediimidedicarboxylic acid, 4,4′-diamino(diphenyl sulfone)diimidedicarboxylic acid, 1,4-naphthalenedicarboxylic acid, 2,6-naphthalenedicarboxylic acid, 1,3-adamantanedicarboxylic acid, 1,8-naphthalenedicarboxylic acid, 2,3-naphthalenedicarboxylic acid, 8-methoxy-2,3-naphthalenedicarboxylic acid, 8-nitro-2,3-naphthalenecarboxylic acid, 8-sulfo-2,3-naphthalenedicarboxylic acid, anthracene-2,3-dicarboxylic acid, 2′,3′-diphenyl-p-ter-phenyl-4,4″-dicarboxylic acid, (diphenyl ether)-4,4′-dicarboxylic acid, imidazole-4,5-di-carboxylic acid, 4(1H)-oxothiochromene-2,8-dicarboxylic acid, 5-tert-butyl-1,3-benzene-dicarboxylic acid, 7,8-quinolinedicarboxylic acid, 4,5-imidazoledicarboxylic acid, 4-

cyclohexene-1,2-dicarboxylic acid, hexatriacontanedicarboxylic acid, tetra-decanedicarboxylic acid, 1,7-heptadicarboxylic acid, 5-hydroxy-1,3-benzenedicarboxylic acid, 2,5-dihydroxy-1,4-dicarboxylic acid, pyrazine-2,3-dicarboxylic acid, furan-2,5-dicarboxylic acid, 1-nonene-6,9-dicarboxylic acid, eicosenedicarboxylic acid, 4,4′-dihydroxydiphenylmethane-3,3′-dicarboxylic acid, 1-amino-4-methyl-9,10-dioxo-9,10-dihydroanthracene-2,3-dicarboxylic acid, 2,5-pyridinedicarboxylic acid, cyclohexene-2,3-dicarboxylic acid, 2,9-dichlorofluorubin-4,11-dicarboxylic acid, 7-chloro-3-methylquinoline-6,8-dicarboxylic acid, 2,4-dichlorobenzophenone-2′,5′-dicarboxylic acid, 1,3-benzenedicarboxylic acid, 2,6-pyridinedicarboxylic acid, 1-methylpyrrole-3,4-dicarboxylic acid, 1-benzyl-1H-pyrrole-3,4-dicarboxylic acid, anthraquinone-1,5-dicarboxylic acid, 3,5-pyrazoledicarboxylic acid, 2-nitrobenzene-1,4-dicarboxylic acid, heptane-1,7-dicarboxylic acid, cyclobutane-1,1-dicarboxylic acid, 1,14-tetradecanedicarboxylic acid, 5,-6-dehydronorbornane-2,3-dicarboxylic acid, 5-ethyl-2,3-pyridinedicarboxylic acid or camphordicarboxylic acid.

[0074] For example, the at least bidentate organic compound can be derived from a tricarboxylic acid that already has at least one isocyanate reactive functional group or that can serve as starting material for the preparation of a tricarboxylic acid by derivatization. Examples of tricarboxylic acids are [0075] 2-hydroxy-1,2,3-propanetricarboxylic acid, 7-chloro-2,3,8-quinolinetricarboxylic acid, 1,2,3-, 1,2,4-benzenetricarboxylic acid, 1,2,4-butanetricarboxylic acid, 2-phosphono-1,2,4-butanetricarboxylic acid, 1,3,5-benzenetricarboxylic acid, 1-hydroxy-1,2,3-propanetricarboxylic acid, 4,5-dihydro-4,5-dioxo-1H-pyrrolo[2,3-F]quinoline-2,7,9-tricarboxylic acid, 5-acetyl-3-amino-6-methylbenzene-1,2,4-tricarboxylic acid, 3-amino-5-benzoyl-6-methylbenzene-1,2,4-tricarboxylic acid, 1,2,3-propanetricarboxylic acid or aurintricarboxylic acid.

[0076] Examples of an at least bidentate organic compound derived from a tetracarboxylic acid that can serve as starting material for the preparation of a tetracarboxylic acid by derivatization, are [0077] 1,1-dioxidoperylo[1,12-BCD]thiophene-3,4,9,10-tetracarboxylic acid, perylenetetracarboxylic acids such as perylene-3,4,9,10-tetracarboxylic acid or (perylene 1,12-sulfone)-3,4,9,10-tetracarboxylic acid, butanetetracarboxylic acids such as 1,2,3,4-butanetetracarboxylic acid or meso-1,2,3,4-butanetetracarboxylic acid, decane-2,4,6,8-tetracarboxylic acid, 1,4,7,10,13,16-hexaoxacyclooctadecane-2,3,11,12-tetracarboxylic acid, 1,2,4,5-benzenetetracarboxylic acid, 1,2,11,12-dodecanetetracarboxylic acid, 1,2,5,6-hexanetetracarboxylic acid, 1,2,7,8-octanetetracarboxylic acid, 1,4,5,8-naphthalenetetracarboxylic acid, 1,2,9,10-decanetetracarboxylic acid, benzophenonetetracarboxylic acid, 3,3′,4,4′-benzophenonetetracarboxylic acid, tetrahydrofurantetracarboxylic acid or cyclopentanetetracarboxylic acids such as cyclopentane-1,2,3,4-tetracarboxylic acid.

[0078] Very particular preference is given to at least monosubstituted aromatic dicarboxylic, tricarboxylic or tetracarboxylic acids having one, two, three, four or more rings, with each of the rings being able to comprise at least one heteroatom and two or more rings being able to comprise identical or different heteroatoms. For example, preference is given to one-ring dicarboxylic acids, one-ring tricarboxylic acids, one-ring tetracarboxylic acids, two-ring dicarboxylic acids, two-ring tricarboxylic acids, two-ring tetracarboxylic acids, three-ring dicarboxylic acids, three-ring tricarboxylic acids, three-ring tetracarboxylic acids, four-ring dicarboxylic acids, four-ring tricarboxylic acids and/or four-ring tetracarboxylic acids. Suitable heteroatoms are, for example, N, O, S, B, P, and preferred heteroatoms are N, S and/or O. Suitable substituents are those isocyanate reactive functional groups mentioned above.

[0079] Especially preferred is metal-organic framework material, wherein the at least bidentate organic compound is an at least monoaminosubstituted aromatic dicarboxylic acid selected from the group consisting of terephthalic acid, 2,6-naphthalenedicarboxylic acid, 1,4-naphthalenedicarboxylic acid, 1,5-naphthalenedicarboxylic acid and derivatives thereof in which at least one CH group in the ring is replaced by nitrogen, wherein the framework in powder form preferably has a specific surface area determined by the Langmuir method (N.sub.2, 77 K) of at

least 2500 m.sup.2/g. Such MOFs are described in WO 2008/142059 A1.
[0080] In a particularly preferred embodiment the at least one at least bidentate organic compound is derived from an amino polycarboxylic acid, preferably derived from an amino dicarboxylic acid, even more preferably derived from 2-aminoterephthalic acid.
[0081] Apart from these at least bidentate organic compounds, the metal-organic framework can also comprise one or more monodentate ligands.
[0082] For the purposes of the present invention, the term “derived” means that the at least one at least bidentate organic compound is present in partially or completely deprotonated form.
[0083] For the purposes of the present invention, the term “derived” also means that the carboxylic acid function can be present as a sulfur analogue. Sulfur analogues are —C(═O)SH and also its tautomer and —C(S)SH.
[0084] Suitable solvents for preparing the metal-organic framework are, inter alia, ethanol, dimethylformamide, toluene, methanol, chlorobenzene, diethylformamide, dimethyl sulfoxide, water, hydrogen peroxide, methylamine, sodium hydroxide solution, N-methylpyrrolidone ether, acetonitrile, benzyl chloride, triethylamine, ethylene glycol and mixtures thereof. Further metal ions, at least bidentate organic compounds and solvents for the preparation of MOFs are described, inter alia, in U.S. Pat. No. 5,648,508 or DE-A 101 11 230.
[0085] The pore size of the metal-organic framework can be controlled by selection of the appropriate ligand and/or the at least bidentate organic compound. In general, the larger the organic compound, the larger the pore size. The pore size is preferably from 0.2 nm to 30 nm, particularly preferably in the range from 0.3 nm to 9 nm, based on the crystalline material.
[0086] However, larger pores whose size distribution can vary also occur in a shaped body comprising a metal-organic framework. Preference is, however, given to more than 50% of the total pore volume, in particular more than 75%, being made up by pores having a pore diameter of up to 1000 mm. However, preference is given to a major part of the pore volume being made up by pores having two diameter ranges. It is therefore preferred for more than 25% of the total pore volume, in particular more than 50% of the total pore volume, to be made up by pores which have a diameter in the range from 100 nm to 800 nm and more than 15% of the total pore volume, in particular more than 25% of the total pore volume, to be made up by pores which have a diameter up to 10 nm. The pore distribution can be determined by means of mercury porosimetry.
[0087] Examples of metal-organic frameworks that already has at least one isocyanate reactive functional group or that can serve as starting material for the preparation of a metal-organic framework material by derivatization are given below. The designation of the framework, the metal and the at least bidentate ligand and also the solvent and the cell parameters (angles α, β and γ and the dimensions A, B and C in Å) are indicated. The latter were determined by X-ray diffraction.
TABLE-US-00001 Constituents molar ratio Space MOF-n M + L Solvents α β γ a b c group MOF-0 Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 120 16.711 16.711 14.189 P6(3)/Mcm H.sub.3(BTC) MOF-2 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 102.8 90 6.718 15.49 12.43 P2(1)/n (0.246 mmol) toluene H.sub.2(BDC) 0.241 mmol) MOF-3 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 99.72 111.11 108.4 9.726 9.911 10.45 P-1 (1.89 mmol) MeOH H.sub.2(BDC) (1.93 mmol) MOF-4 Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 90 14.728 14.728 14.728 P2(1)3 (1.00 mmol) H.sub.3(BTC) (0.5 mmol) MOF-5 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 90 90 25.669 25.669 25.669 Fm-3m (2.22 mmol) chlorobenzene H.sub.2(BDC) (2.17 mmol) MOF-38 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 90 90 20.657 20.657 17.84 14cm (0.27 mmol) chlorobenzene H.sub.3(BTC) (0.15 mmol) MOF-31 Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 90 10.821 10.821 10.821 Pn(-3)m Zn(ADC).sub.2 0.4 mmol H.sub.2(ADC) 0.8 mmol MOF-12 Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 90 15.745 16.907 18.167 Pbca Zn.sub.2(ATC) 0.3 mmol H.sub.4(ATC) 0.15 mmol MOF-20 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 92.13 90 8.13 16.444 12.807 P2(1)/c ZnNDC 0.37 mmol chlorobenzene H.sub.2NDC 0.36 mmol MOF-37 Zn(NO.sub.3).sub.2•6H.sub.2O DEF 72.38 83.16 84.33 9.952 11.576 15.556 P-1 0.2 mmol chloro-

H.sub.2NDC benzene 0.2 mmol MOF-8 Tb(NO.sub.3).sub.3•5H.sub.2O DMSO 90 115.7 90 19.83 9.822 19.183 C2/c Tb.sub.2 (ADC) 0.10 mmol MeOH H.sub.2ADC 0.20 mmol MOF-9 Tb(NO.sub.3).sub.3•5H.sub.2O DMSO 90 102.09 90 27.056 16.795 28.139 C2/c Tb.sub.2 (ADC) 0.08 mmol H.sub.2ADB 0.12 mmol MOF-6 Tb(NO.sub.3).sub.3•5H.sub.2O DMF 90 91.28 90 17.599 19.996 10.545 P21/c 0.30 mmol MeOH H.sub.2 (BDC) 0.30 mmol MOF-7 Tb(NO.sub.3).sub.3•5H.sub.2O H.sub.2O 102.3 91.12 101.5 6.142 10.069 10.096 P-1 0.15 mmol H.sub.2(BDC) 0.15 mmol MOF-69A Zn(NO.sub.3).sub.2•6H.sub.2O DEF 90 111.6 90 23.12 20.92 12 C2/c 0.083 mmol H.sub.2O.sub.2 4,4′BPDC MeNH.sub.2 0.041 mmol MOF-69B Zn(NO.sub.3).sub.2•6H.sub.2O DEF 90 95.3 90 20.17 18.55 12.16 C2/c 0.083 mmol H.sub.2O.sub.2 2,6-NCD MeNH.sub.2 0.041 mmol MOF-11 Cu(NO.sub.3).sub.2•2.5H.sub.2O H.sub.2O 90 93.86 90 12.987 11.22 11.336 C2/c Cu.sub.2(ATC) 0.47 mmol H.sub.2ATC 0.22 mmol MOF-11 90 90 90 8.4671 8.4671 14.44 P42/mmc Cu.sub.2(ATC) dehydr. MOF-14 Cu(NO.sub.3).sub.2•2.5H.sub.2O H.sub.2O 90 90 90 26.946 26.946 26.946 Im-3 Cu.sub.3 (BTB) 0.28 mmol DMF H.sub.3BTB EtOH 0.052 mmol MOF-32 Cd(NO.sub.3).sub.2•4H.sub.2O H.sub.2O 90 90 90 13.468 13.468 13.468 P(-4)3m Cd(ATC) 0.24 mmol NaOH H.sub.4ATC 0.10 mmol MOF-33 ZnCl.sub.2 H.sub.2O 90 90 90 19.561 15.255 23.404 Imma Zn.sub.2 (ATB) 0.15 mmol DMF H.sub.4ATB EtOH 0.02 mmol MOF-34 Ni(NO.sub.3).sub.2•6H.sub.2O H.sub.2O 90 90 90 10.066 11.163 19.201 P2.sub.12.sub.12.sub.1 Ni(ATC) 0.24 mmol NaOH H.sub.4ATC 0.10 mmol MOF-36 Zn(NO.sub.3).sub.2•4H.sub.2O H.sub.2O 90 90 90 15.745 16.907 18.167 Pbca Zn.sub.2 (MTB) 0.20 mmol DMF H.sub.4MTB 0.04 mmol MOF-39 Zn(NO.sub.3).sub.2 4H.sub.2O H.sub.2O 90 90 90 17.158 21.591 25.308 Pnma Zn.sub.3O(HBTB) 0.27 mmol DMF H.sub.3BTB EtOH 0.07 mmol NO305 FeCl.sub.2•4H.sub.2O DMF 90 90 120 8.2692 8.2692 63.566 R-3c 5.03 mmol formic acid 86.90 mmol NO306A FeCl.sub.2•4H.sub.2O DEF 90 90 90 9.9364 18.374 18.374 Pbcn 5.03 mmol formic acid 86.90 mmol NO29 Mn(Ac).sub.2•4H.sub.2O DMF 120 90 90 14.16 33.521 33.521 P-1 MOF-0 0.46 mmol similar H.sub.3BTC 0.69 mmol BPR48 Zn(NO.sub.3).sub.2 6H.sub.2O DMSO 90 90 90 14.5 17.04 18.02 Pbca A2 0.012 mmol toluene H.sub.2BDC 0.012 mmol BPR69 Cd(NO.sub.3).sub.2 4H.sub.2O DMSO 90 98.76 90 14.16 15.72 17.66 Cc B1 0.0212 mmol H.sub.2BDC 0.0428 mmol BPR92 Co(NO.sub.3).sub.2•6H.sub.2O NMP 106.3 107.63 107.2 7.5308 10.942 11.025 P1 A2 0.018 mmol H.sub.2BDC 0.018 mmol BPR95 Cd(NO.sub.3).sub.2 4H.sub.2O NMP 90 112.8 90 14.460 11.085 15.829 P2(1)/n C5 0.012 mmol H.sub.2BDC 0.36 mmol Cu C.sub.6H.sub.4O.sub.6 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 105.29 90 15.259 14.816 14.13 P2(1)/c 0.370 mmol chlorobenzene H.sub.2BDC(OH).sub.2 0.37 mmol M(BTC) Co(SO.sub.4) H.sub.2O DMF as for MOF-0 MOF-0 0.055 mmol similar H.sub.3BTC 0.037 mmol Tb(C.sub.6H.sub.4O.sub.6) Tb(NO.sub.3).sub.3•5H.sub.2O DMF 104.6 107.9 97.147 10.491 10.981 12.541 P-1 0.370 mmol chlorobenzene H.sub.2(C.sub.6H.sub.4O.sub.6) 0.56 mmol Zn(C.sub.2O.sub.4) ZnCl.sub.2 DMF 90 120 90 9.4168 9.4168 8.464 P(-3)1m 0.370 mmol chlorobenzene oxalic acid 0.37 mmol Co(CHO) Co(NO.sub.3).sub.2•5H.sub.2O DMF 90 91.32 90 11.328 10.049 14.854 P2(1)/n 0.043 mmol formic acid 1.60 mmol Cd(CHO) Cd(NO.sub.3).sub.2•4H.sub.2O DMF 90 120 90 8.5168 8.5168 22.674 R-3c 0.185 mmol formic acid 0.185 mmol Cu(C.sub.3H.sub.2O.sub.4) Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 90 90 8.366 8.366 11.919 P43 0.043 mmol malonic acid 0.192 mmol Zn.sub.6 (NDC).sub.5 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 95.902 90 19.504 16.482 14.64 C2/m MOF-48 0.097 mmol chlorobenzene 14 NDC H.sub.2O.sub.2 0.069 mmol MOF-47 Zn(NO.sub.3).sub.2 6H.sub.2O DMF 90 92.55 90 11.303 16.029 17.535 P2(1)/c 0.185 mmol chlorobenzene H.sub.2(BDC[CH.sub.3].sub.4) H.sub.2O.sub.2 0.185 mmol MO25 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 112.0 90 23.880 16.834 18.389 P2(1)/c 0.084 mmol BPhDC 0.085 mmol Cu-Thio Cu(NO.sub.3).sub.2•2.5H.sub.2O DEF 90 113.6 90 15.4747 14.514 14.032 P2(1)/c 0.084 mmol thiophenedicarboxylic acid 0.085 mmol CIBDC1 Cu(NO.sub.3).sub.2•2.5H.sub.2O0.0 DMF 90 105.6 90 14.911 15.622 18.413 C2/c 84 mmol H.sub.2(BDCCl.sub.2) 0.085 mmol MOF-101 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 90 90

21.607 20.607 20.073 Fm3m 0.084 mmol BrBDC 0.085 mmol Zn.sub.3(BTC).sub.2 ZnCl.sub.2 DMF 90 90 90 26.572 26.572 26.572 Fm-3m 0.033 mmol EtOH H.sub.3BTC base 0.033 mmol added MOF-j Co(CH.sub.3CO.sub.2).sub.2•4H.sub.2O H.sub.2O 90 112.0 90 17.482 12.963 6.559 C2 (1.65 mmol) H.sub.3(BZC) (0.95 mmol) MOF-n Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 120 16.711 16.711 14.189 P6(3)/mcm H.sub.3 (BTC) PbBDC Pb(NO.sub.3).sub.2 DMF 90 102.7 90 8.3639 17.991 9.9617 P2(1)/n (0.181 mmol) ethanol H.sub.2(BDC) (0.181 mmol) Znhex Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 90 120 37.1165 37.117 30.019 P3(1)c (0.171 mmol) p-xylene H.sub.3BTB ethanol (0.114 mmol) AS16 FeBr.sub.2 DMF 90 90.13 90 7.2595 8.7894 19.484 P2(1)c 0.927 mmol anhydr. H.sub.2(BDC) 0.927 mmol AS27-2 FeBr.sub.2 DMF 90 90 90 26.735 26.735 26.735 Fm3m 0.927 mmol anhydr. H.sub.3(BDC) 0.464 mmol AS32 FeCl.sub.3 DMF 90 90 120 12.535 12.535 18.479 P6(2)c 1.23 mmol anhydr. H.sub.2(BDC) ethanol 1.23 mmol AS54-3 FeBr.sub.2 DMF 90 109.98 90 12.019 15.286 14.399 C2 0.927 anhydr. BPDC n-0.927 mmol propanol AS61-4 FeBr.sub.2 pyridine 90 90 120 13.017 13.017 14.896 P6(2)c 0.927 mmol anhydr. m-BDC 0.927 mmol AS68-7 FeBr.sub.2 DMF 90 90 90 18.3407 10.036 18.039 Pca2.sub.1 0.927 mmol anhydr. m-BDC pyridine 1.204 mmol Zn(ADC) Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 99.85 90 16.764 9.349 9.635 C2/c 0.37 mmol chlorobenzene H.sub.2(ADC) 0.36 mmol MOF-12 Zn(NO.sub.3).sub.2•6H.sub.2O ethanol 90 90 90 15.745 16.907 18.167 Pbca Zn.sub.2 (ATC) 0.30 mmol H.sub.4(ATC) 0.15 mmol MOF-20 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 92.13 90 8.13 16.444 12.807 P2(1)/c ZnNDC 0.37 mmol chlorobenzene H.sub.2NDC 0.36 mmol MOF-37 Zn(NO.sub.3).sub.2•6H.sub.2O DEF 72.38 83.16 84.33 9.952 11.576 15.556 P-1 0.20 mmol chlorobenzene H.sub.2NDC 0.20 mmol Zn(NDC) Zn(NO.sub.3).sub.2•6H.sub.2O DMSO 68.08 75.33 88.31 8.631 10.207 13.114 P-1 (DMSO) H.sub.2NDC Zn(NDC) Zn(NO.sub.3).sub.2•6H.sub.2O 90 99.2 90 19.289 17.628 15.052 C2/c H.sub.2NDC Zn(HPDC) Zn(NO.sub.3).sub.2•4H.sub.2O DMF 107.9 105.06 94.4 8.326 12.085 13.767 P-1 0.23 mmol H2O H.sub.2(HPDC) 0.05 mmol Co(HPDC) Co(NO.sub.3).sub.2•6H.sub.2O DMF 90 97.69 90 29.677 9.63 7.981 C2/c 0.21 mmol H.sub.2O/ethanol H.sub.2 (HPDC) 0.06 mmol Zn.sub.3(PDC)2.5 Zn(NO.sub.3).sub.2•4H.sub.2O DMF/CIBz 79.34 80.8 85.83 8.564 14.046 26.428 P-1 0.17 mmol H.sub.2O/TEA H.sub.2(HPDC) 0.05 mmol Cd.sub.2 (TPDC)2 Cd(NO.sub.3).sub.2•4H.sub.2O methanol/CHP 70.59 72.75 87.14 10.102 14.412 14.964 P-1 0.06 mmol H.sub.2O H.sub.2(HPDC) 0.06 mmol Tb(PDC)1.5 Tb(NO.sub.3).sub.3•5H.sub.2O DMF 109.8 103.61 100.14 9.829 12.11 14.628 P-1 0.21 mmol H.sub.2O/ethanol H.sub.2(PDC) 0.034 mmol ZnDBP Zn(NO.sub.3).sub.2•6H.sub.2O MeOH 90 93.67 90 9.254 10.762 27.93 P2/n 0.05 mmol dibenzyl phosphate 0.10 mmol Zn.sub.3(BPDC) ZnBr.sub.2 DMF 90 102.76 90 11.49 14.79 19.18 P21/n 0.021 mmol 4,4′BPDC 0.005 mmol CdBDC Cd(NO.sub.3).sub.2•4H.sub.2O DMF 90 95.85 90 11.2 11.11 16.71 P21/n 0.100 mmol Na.sub.2SiO.sub.3 H.sub.2(BDC) (aq) 0.401 mmol Cd-mBDC Cd(NO.sub.3).sub.2•4H.sub.2O DMF 90 101.1 90 13.69 18.25 14.91 C2/c 0.009 mmol MeNH.sub.2 H.sub.2(mBDC) 0.018 mmol Zn.sub.4OBNDC Zn(NO.sub.3).sub.2•6H.sub.2O DEF 90 90 90 22.35 26.05 59.56 Fmmm 0.041 mmol MeNH.sub.2 BNDC H.sub.2O.sub.2 Eu(TCA) Eu(NO.sub.3).sub.3•6H.sub.2O DMF 90 90 90 23.325 23.325 23.325 Pm-3n 0.14 mmol chlorobenzene TCA 0.026 mmol Tb(TCA) Tb(NO.sub.3).sub.3•6H.sub.2O DMF 90 90 90 23.272 23.272 23.372 Pm-3n 0.069 mmol chlorobenzene TCA 0.026 mmol Formates Ce(NO.sub.3).sub.3•6H.sub.2O H.sub.2O 90 90 120 10.668 10.667 4.107 R-3m 0.138 mmol ethanol formic acid 0.43 mmol FeCl.sub.2•4H.sub.2O DMF 90 90 120 8.2692 8.2692 63.566 R-3c 5.03 mmol formic acid 86.90 mmol FeCl.sub.2•4H.sub.2O DEF 90 90 90 9.9364 18.374 18.374 Pbcn 5.03 mmol formic acid 86.90 mmol FeCl.sub.2•4H.sub.2O DEF 90 90 90 8.335 8.335 13.34 P-31c 5.03 mmol formic acid 86.90 mmol NO330 FeCl.sub.2•4H.sub.2O form- 90 90 90 8.7749 11.655 8.3297 Pnna 0.50 mmol amide formic acid 8.69 mmol NO332 FeCl.sub.2•4H.sub.2O DIP 90 90 90 10.0313 18.808 18.355 Pbcn 0.50 mmol formic acid 8.69 mmol NO333 FeCl.sub.2•4H.sub.2O DBF 90 90 90 45.2754 23.861 12.441 Cmcm 0.50 mmol formic acid 8.69 mmol NO335 FeCl.sub.2•4H.sub.2O CHF 90 91.372 90

11.5964 10.187 14.945 P21/n 0.50 mmol formic acid 8.69 mmol NO336 FeCl.sub.2•4H.sub.2O MFA 90 90 90 11.7945 48.843 8.4136 Pbcm 0.50 mmol formic acid 8.69 mmol NO13 Mn(Ac).sub.2•4H.sub.2O ethanol 90 90 90 18.66 11.762 9.418 Pbcn 0.46 mmol benzoic acid 0.92 mmol bipyridine 0.46 mmol NO29 Mn(Ac).sub.2•4H.sub.2O DMF 120 90 90 14.16 33.521 33.521 P-1 MOF-0 0.46 mmol similar H.sub.3BTC 0.69 mmol Mn(hfac).sub.2 Mn(Ac).sub.2•4H.sub.2O ether 90 95.32 90 9.572 17.162 14.041 C2/c (O.sub.2CC.sub.6H.sub.5) 0.46 mmol Hfac 0.92 mmol bipyridine 0.46 mmol BPR43G2 Zn(NO.sub.3).sub.2•6H.sub.2O DMF 90 91.37 90 17.96 6.38 7.19 C2/c 0.0288 mmol CH.sub.3CN H.sub.2BDC 0.0072 mmol BPR48A2 Zn(NO.sub.3).sub.2 6H.sub.2O DMSO 90 90 90 14.5 17.04 18.02 Pbca 0.012 mmol toluene H.sub.2BDC 0.012 mmol BPR49B1 Zn(NO.sub.3).sub.2 6H.sub.2O DMSC 90 91.172 90 33.181 9.824 17.884 C2/c 0.024 mmol methanol H.sub.2BDC 0.048 mmol BPR56E1 Zn(NO.sub.3).sub.2 6H.sub.2O DMSO 90 90.096 90 14.5873 14.153 17.183 P2(1)/n 0.012 mmol n- H.sub.2BDC propanol 0.024 mmol BPR68D10 Zn(NO.sub.3).sub.2 6H.sub.2O DMSO 90 95.316 90 10.0627 10.17 16.413 P2(1)/c 0.0016 mmol benzene H.sub.3BTC 0.0064 mmol BPR69B1 Cd(NO.sub.3).sub.2 4H.sub.2O DMSO 90 98.76 90 14.16 15.72 17.66 Cc 0.0212 mmol H.sub.2BDC 0.0428 mmol BPR73E4 Cd(NO.sub.3).sub.2 4H.sub.2O DMSO 90 92.324 90 8.7231 7.0568 18.438 P2(1)/n 0.006 mmol toluene H.sub.2BDC 0.003 mmol BPR76D5 Zn(NO.sub.3).sub.2 6H.sub.2O DMSO 90 104.17 90 14.4191 6.2599 7.0611 Pc 0.0009 mmol H.sub.2BzPDC 0.0036 mmol BPR80B5 Cd(NO.sub.3).sub.2•4H.sub.2O DMF 90 115.11 90 28.049 9.184 17.837 C2/c 0.018 mmol H.sub.2BDC 0.036 mmol BPR80H5 Cd(NO.sub.3).sub.2 4H.sub.2O DMF 90 119.06 90 11.4746 6.2151 17.268 P2/c 0.027 mmol H.sub.2BDC 0.027 mmol BPR82C6 Cd(NO.sub.3).sub.2 4H.sub.2O DMF 90 90 90 9.7721 21.142 27.77 Fdd2 0.0068 mmol H.sub.2BDC 0.202 mmol BPR86C3 Co(NO.sub.3).sub.2 6H.sub.2O DMF 90 90 90 18.3449 10.031 17.983 Pca2(1) 0.0025 mmol H.sub.2BDC 0.075 mmol BPR86H6 Cd(NO.sub.3).sub.2•6H.sub.2O DMF 80.98 89.69 83.412 9.8752 10.263 15.362 P-1 0.010 mmol H.sub.2BDC 0.010 mmol Co(NO.sub.3).sub.2 6H.sub.2O NMP 106.3 107.63 107.2 7.5308 10.942 11.025 P1 BPR95A2 Zn(NO.sub.3).sub.2 6H.sub.2O NMP 90 102.9 90 7.4502 13.767 12.713 P2(1)/c 0.012 mmol H.sub.2BDC 0.012 mmol CuC.sub.6F.sub.4O.sub.4 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 98.834 90 10.9675 24.43 22.553 P2(1)/n 0.370 mmol chlorobenzene H.sub.2BDC(OH).sub.2 0.37 mmol Fe formic FeCl.sub.2•4H.sub.2O DMF 90 91.543 90 11.495 9.963 14.48 P2(1)/n 0.370 mmol formic acid 0.37 mmol Mg formic Mg(NO.sub.3).sub.2•6H.sub.2O DMF 90 91.359 90 11.383 9.932 14.656 P2(1)/n 0.370 mmol formic acid 0.37 mmol MgC.sub.6H.sub.4O.sub.6 Mg(NO.sub.3).sub.2•6H.sub.2O DMF 90 96.624 90 17.245 9.943 9.273 C2/c 0.370 mmol H.sub.2BDC(OH).sub.2 0.37 mmol Zn C.sub.2H.sub.4BDC ZnCl.sub.2 DMF 90 94.714 90 7.3386 16.834 12.52 P2(1)/n MOF-38 0.44 mmol CBBDC 0.261 mmol MOF-49 ZnCl.sub.2 DMF 90 93.459 90 13.509 11.984 27.039 P2/c 0.44 mmol CH.sub.3CN m-BDC 0.261 mmol MOF-26 Cu(NO.sub.3).sub.2•5H.sub.2O DMF 90 95.607 90 20.8797 16.017 26.176 P2(1)/n 0.084 mmol DCPE 0.085 mmol MOF-112 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 107.49 90 29.3241 21.297 18.069 C2/c 0.084 mmol ethanol o-Br-m-BDC 0.085 mmol MOF-109 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 111.98 90 23.8801 16.834 18.389 P2(1)/c 0.084 mmol KDB 0.085 mmol MOF-111 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 102.16 90 10.6767 18.781 21.052 C2/c 0.084 mmol ethanol o-BrBDC 0.085 mmol MOF-110 Cu(NO.sub.3).sub.2•2.5H.sub.2O DMF 90 90 120 20.0652 20.065 20.747 R-3/m 0.084 mmol thiophenedicarboxylic acid 0.085 mmol MOF-107 Cu(NO.sub.3).sub.2•2.5H.sub.2O DEF 104.8 97.075 95.206 11.032 18.067 18.452 P-1 0.084 mmol thiophenedicarboxy lic acid 0.085 mmol MOF-108 Cu(NO3)2•2.5H2O DBF/methanol 90 113.63 90 15.4747 14.514 14.032 C2/c 0.084 mmol thiophenedicarboxylic acid 0.085 mmol MOF-102 Cu(NO3)2•2.5H2O DMF 91.63 106.24 112.01 9.3845 10.794 10.831 P-1 0.084 mmol H2(BDCCl2) 0.085 mmol Clbdc1 Cu(NO3)2•2.5H2O DEF 90 105.56 90 14.911 15.622 18.413 P-1 0.084 mmol H2(BDCCl2) 0.085 mmol Cu(NMOP) Cu(NO3)2•2.5H2O DMF 90 102.37 90 14.9238 18.727 15.529 P2(1)/m 0.084

mmol NBDC 0.085 mmol Tb(BTC) Tb(NO3)3•5H2O DMF 90 106.02 90 18.6986 11.368 19.721 0.033 mmol H3BTC 0.033 mmol Zn3(BTC)2 ZnCl2 DMF 90 90 90 26.572 26.572 26.572 Fm-3m Honk 0.033 mmol ethanol H3BTC 0.033 mmol Zn4O(NDC) Zn(NO3)2•4H2O DMF 90 90 90 41.5594 18.818 17.574 aba2 0.066 mmol ethanol 14NDC 0.066 mmol CdTDC Cd(NO3)2•4H2O DMF 90 90 90 12.173 10.485 7.33 Pmma 0.014 mmol H2O thiophene 0.040 mmol DABCO 0.020 mmol IRMOF-2 Zn(NO3)2•4H2O DEF 90 90 90 25.772 25.772 25.772 Fm-3m 0.160 mmol o-Br-BDC 0.60 mmol IRMOF-3 Zn(NO3)2•4H2O DEF 90 90 90 25.747 25.747 25.747 Fm-3m 0.20 mmol ethanol H2N-BDC 0.60 mmol IRMOF-4 Zn(NO3)2•4H2O DEF 90 90 90 25.849 25.849 25.849 Fm-3m 0.11 mmol [C3H7O]2-BDC 0.48 mmol IRMOF-5 Zn(NO3)2•4H2O DEF 90 90 90 12.882 12.882 12.882 Pm-3m 0.13 mmol [C5H11O]2-BDC 0.50 mmol IRMOF-6 Zn(NO3)2•4H2O DEF 90 90 90 25.842 25.842 25.842 Fm-3m 0.20 mmol [C2H4]-BDC 0.60 mmol IRMOF-7 Zn(NO3)2•4H2O DEF 90 90 90 12.914 12.914 12.914 Pm-3m 0.07 mmol 1,4NDC 0.20 mmol IRMOF-8 Zn(NO3)2•4H2O DEF 90 90 90 30.092 30.092 30.092 Fm-3m 0.55 mmol 2,6NDC 0.42 mmol IRMOF-9 Zn(NO3)2•4H2O DEF 90 90 90 17.147 23.322 25.255 Pnnm 0.05 mmol BPDC 0.42 mmol IRMOF-10 Zn(NO3)2•4H2O DEF 90 90 90 34.281 34.281 34.281 Fm-3m 0.02 mmol BPDC 0.012 mmol IRMOF-11 Zn(NO3)2•4H2O DEF 90 90 90 24.822 24.822 56.734 R-3m 0.05 mmol HPDC 0.20 mmol IRMOF-12 Zn(NO3)2•4H2O DEF 90 90 90 34.281 34.281 34.281 Fm-3m 0.017 mmol HPDC 0.12 mmol IRMOF-13 Zn(NO.sub.3).sub.2•4H.sub.2O DEF 90 90 90 24.822 24.822 56.734 R-3m 0.048 mmol PDC 0.31 mmol IRMOF-14 Zn(NO.sub.3).sub.2•4H.sub.2O DEF 90 90 90 34.381 34.381 34.381 Fm-3m 0.17 mmol PDC 0.12 mmol IRMOF-15 Zn(NO.sub.3).sub.2•4H.sub.2O DEF 90 90 90 21.459 21.459 21.459 Im-3m 0.063 mmol TPDC 0.025 mmol IRMOF-16 Zn(NO.sub.3).sub.2•4H.sub.2O DEF 90 90 90 21.49 21.49 21.49 Pm-3m 0.0126 mmol NMP TPDC 0.05 mmol ADC Acetylenedicarboxylic acid NDC Naphthalenedicarboxylic acid BDC Benzenedicarboxylic acid ATC Adamantanetetracarboxylic acid BTC Benzenetricarboxylic acid BTB Benzenetribenzoic acid MTB Methanetetrabenzoic acid ATB Adamantanetetrabenzoic acid ADB Adamantanedibenzoic acid

[0088] Further metal-organic frameworks are MOF-2 to 4, MOF-9, MOF-31 to 36, MOF-39, MOF-69 to 80, MOF103 to 106, MOF-122, MOF-125, MOF-150, MOF-177, MOF-178, MOF-235, MOF-236, MOF-500, MOF-501, MOF-502, MOF-505, IRMOF-1, IRMOF-61, IRMOP-13, IRMOP-51, MIL-17, MIL-45, MIL-47, MIL-53, MIL-59, MIL-60, MIL-61, MIL-63, MIL-68, MIL-79, MIL-80, MIL-83, MIL-85, CPL-1 to 2, SZL-1, which are described in the literature.

[0089] Particularly preferred metal-organic frameworks are MIL-53, Zn-tBu-isophthalic acid, Al-BDC, MOF-5, MOF-177, MOF-505, IRMOF-8, IRMOF-11, Cu-BTC, Al-NDC, Al-aminoBDC, Cu-BDC-TEDA, Zn-BDC-TEDA, Al-BTC, Cu-BTC, Al-NDC, Mg-NDC, Al-fumarate, Zn-2-methylimidazolate, Zn-2-aminoimidazolate, Cu-biphenyldicarboxylate-TEDA, MOF-74, Cu-BPP, Sc-terephthalate. Greater preference is given to Sc-terephthalate, Cu-BTC, Al-BDC, Al-aminoBDC and Al-BTC. Very particular preference is given to Al-aminoBDC.

[0090] Al-amino BDC (Al-2-aminoterephthalic acid MOF) can be prepared according to WO 2008/142059 A1.

[0091] Apart from the conventional method of preparing the MOFs, as described, for example, in U.S. Pat. No. 5,648,508, these can also be prepared by an electrochemical route. In this respect, reference is made to DE-A 103 55 087 and WO-A 2005/049892. The metal-organic frameworks prepared in this way have particularly good properties with regard to the adsorption and desorption of chemical substances, in particular gases.

[0092] Regardless of the method of preparation, the metal-organic framework is generally obtained in pulverulent or crystalline form. The metal-organic framework can also be converted into a shaped body and used in this form according to the present invention.

[0093] The contacting of the polyurethane prepolymer with the porous material can be carried out by simple mixing. No particular requirements are associated with the contacting. The porous material can be dried before contacting. Preferably, conditions for allowing chemical reaction with

the residual monomeric isocyanate are standard conditions, i.e. no elevated pressure, stirring or shaking and room temperature or elevated temperatures. Thus, it is known that amino groups spontaneously react with the isocyanate functional group to yield a urea group.
[0094] The polyurethane prepolymer according to the present invention has the same or very similar characteristics properties as the prepolymer before contacting with the porous material and can be used in the same fields mentioned above.
[0095] However, it is also possible to remove the porous material after contacting by separation, e.g. filtration. Optionally, a second and further contacting with porous material after separation is possible in order to further remove residual monomeric isocyanate.
[0096] The following examples further illustrate the invention.

Description

EXAMPLES
1. Raw Materials
Isocyanate Prepolymer 1
[0097] The prepolymer was prepared by charging 261 g of toluene diisocyanate (TDI) with a ratio of 2,4-TDI:2,6-TDI=80:20 and 0.1 g Diglycol-bis-chloroformiat (DIBIS) into a stirred four-necked flask under nitrogen blanket, heating to 60° C. and slowly adding 739 g of poly(tetramethylene ether glycol) with an OH value of 112 mg KOH/g. The reaction mixture was stirred for 2 hours at 80° C. After cooling down to room temperature, the isocyanate prepolymer was stored at room temperature in a sealed PE bottle free of moisture. NCO content was 6.09%, free monomer content was 0.4% 2,4-TDI and 2.1% 2,6-TDI and the viscosity was determined as 11.4 Pas at 25° C.
Isocyanate Prepolymer 2
[0098] The prepolymer was prepared by charging 320 g of methylene diphenyl diisocyanate (MDI) with a ratio of 4,4′-MDI:2,4′-MDI:2,2′-MDI=1:1:0.05 and 0.1 g Diglycol-bis-chloroformiat (DIBIS) into a stirred four-necked flask under nitrogen blanket, heating to 60° C. and adding 679.9 g of poly (propylene glycol) with an OH value of 104 mg KOH/g. The reaction mixture was stirred for 2 hours at 80° C. After cooling down to room temperature, the isocyanate prepolymer was stored at room temperature in a sealed PE bottle free of moisture. NCO content was 5.27%, free monomer content was 2.1% 4,4′-MDI, 4.7% 2,4′-MDI and 0.4% 2,2′-MDI and the viscosity was determined as 2.8 Pas at 60° C.
Basolite A 100
[0099] Metal-organic framework (MOF) based of aluminum terephthalate with MIL-53 structure from BASF.
Basolite A 120
[0100] Metal-organic framework (MOF) based of aluminum 2-aminoterephthalate prepared according to example 1 of WO 2008/142059 A1.
Aerosil R 504
[0101] Fumed silica from EVONIK with (3-aminopropyl)trimethoxysilane surface functionalization and 125-175 m.sup.2/g BET surface area.
Omyacarb 2T-AV
[0102] Fine chalk powder from OMYA, composition is 97,5% calcium carbonate and <2% magnesium carbonate.
[0103] 2-Ethyl-1-hexanol: Purity >99% from Sigma-Aldrich.
Purolite A110
[0104] Ion exchange resin from PUROLITE based on spherical beads (300-1200 μm) of microporous polystyrene crosslinked with divinylbenzene. Surface functionalized with primary amines.

Lewatit VP OC 1065

[0105] Ion exchange resin from LANXESS based on a microporous polystyrene crosslinked with divinylbenzene in spherical bead form (300-1200 μm). Surface functionalized with benzylamine groups. 50 m.sup.2/g BET surface.

Aerosil VPR 511

[0106] Fumed silica from EVONIK with (3-aminopropyl)trimethoxysilane surface functionalization, 125-175 m.sup.2/g BET surface area.

Mg-Dihydroxyterephthalate

[0107] 12 g of 2,5-dihydroxyterephthalic acid has been dissolved in 685 g N,N-dimethylformamide. To this solution 53.3 g magnesiumacetate-tetrahydrate have been added. The solution has been stirred for 2 hours at 20° C. The solid was filtered off and washed with 150 ml N,N-dimethylformamide and 150 ml methanol, followed by 16 hours of hot extraction with methanol. The material was dried at 130° C. for 16 hours. The BET surface area was determined with nitrogen adsorption and was found to be 1126 m.sup.2/g.

Mg-Hydroxyterephthalate

[0108] 10 g of 2-hydroxyterephthalic acid has been dissolved in 634 g N,N-dimethylformamide. To this solution 49.3 g magnesiumacetate-tetrahydrate have been added. The solution has been stirred for 2 hours at 20° C. The solid was filtered off and washed with 100 ml N,N-dimethylformamide and 100 ml methanol, followed by 16 hours of hot extraction with methanol. The material was dried at 130° C. for 16 hours. The BET surface area was determined with nitrogen adsorption and was found to be 45.5 m.sup.2/g.

Al-Dihydroxyterephthalate

[0109] 12.24 g of 2,5-dihydroxyterephthalic acid has been dissolved in 698.5 g N,N-dimethylformamide. To this solution 39.2 g basic aluminum-diacetate have been added. The solution has been stirred for 2 hours at 20° C. The solid was filtered off and washed with 150 ml N, N-dimethylformamide and 150 ml methanol, followed by hot extraction with methanol for 16 hours. The material was dried at 130° C. for 16 hours. The BET surface area was determined with nitrogen adsorption and was found to be 364 m.sup.2/g.

Al-Hydroxyterphthalate

[0110] 9.5 g of 2-hydroxyterephthalic acid has been dissolved in 600 g N,N-dimethylformamide. To this solution 33.8 g basic aluminum-diacetate have been added. The solution has been stirred for 2 hours at 20° C. The solid was filtered off and washed with 100 ml N,N-dimethylformamide and 100 ml methanol, followed by 16 hours of hot extraction with methanol. The material was dried at 130° C. for 16 hours. The BET surface area was determined with nitrogen adsorption and was found to be 332 m.sup.2/g.

2. Analytical Methods

[0111] The NCO content of prepolymers and prepolymer/absorbent mixtures was determined by backtitration of the respective samples derivatized with 1 molar n-butylamine solution in chlorobenzene using 1 molar hydrochloric acid.

[0112] The free isocyanate monomer content of prepolymers and prepolymer/absorbent mixtures was measured using high performance liquid chromatography (HPLC) with UV detection of samples derivatized with 1-(2-Piridyl)piperazine. This method was applied for MDI and TDI monomer determination.

[0113] The viscosity was determined with a Haake-Mars rheometer with plate-plate geometry in rotation and 1 mm gap width at a shear rate of 10 and 40 s.sup.−1 at the given temperature.

3. Working Examples

[0114] All absorbents were dried under vacuo at 80° C. overnight before use. Lewatit VPOC 1065, Purolite A110 and Aerosil VPR 511 were dried under vacuo at 80° C. for an extended time of 3 days. The residual content of volatiles (mainly water) was determined gravimetrically with Sartorius MA30 moisture analyzed by heating the samples for 10 min at 120° C. Residual moisture was 1.7% for Basolite A100, 2.6% for Basolite A120, 1.3% for Aerosil R 504, 0.1% for Omyacarb 2T-AV, 0.6% for Purolite A110, 0,8% for Lewatit VPOC 1065, 1.1% for Aerosil VPR 511, 4.0% for Mg-DHT, 4.3% for Mg-HT, 1.1% for Al-DHT and 1.8% for Al-HT. After drying the absorbents were directly used for the prepolymer/absorbent mixtures. Typical preparation procedure: 47.5 g of isocyanate prepolymer was charged into a PE cup, closed with a lid and heated to 60° C. in a laboratory heating oven. 2.5 g of dried absorbent was added and homogenized with a Vollrath laboratory mixer. The mixtures with isocyanate prepolymer 1 were stored at 60° C. for 1 hour, those with isocyanate prepolymer 2 at 60° C. for 2 hours. After the thermal annealing the samples were stored in sealed glass vials at room temperature for another 24 hours. NCO content, viscosity and monomer content of the mixtures was determined.

[0115] The examples 1-12 were prepared following the above mentioned process and demonstrate the individual ability of TDI monomer capture for various absorbents. As shown in Table 1 the NCO values of the examples 1-12 decrease in the mixtures, from 6.09% of the untreated isocyanate prepolymer 1 to 4.47% as minimum. The initial TDI monomer content of 2.5% was reduced in all examples 1-12, however, only example 2 containing Basolite A 120 achieved a monomer content as low as 0.05%. This corresponds to a capture of 98% of the free TDI monomers and is significantly improved compared to the examples 1 and 3-12. At the same time the viscosity increased only moderately in example 2 from initially 11.4 (20° C.) Pas to 22.7 Pas (20° C.). The best performing comparison example 8, which achieved a TDI monomer content of 0.72%, exhibits a viscosity of 54.5 Pas (20° C.). These results demonstrate that Basolite A 120 very efficiently captures the TDI monomers to yield prepolymer mixtures with lower than 0.1% residual monomer while keeping the viscosity increase on a moderate level.

TABLE-US-00002 TABLE 1 Examples 1-12 using isocyanate prepolymer 1 based on TDI. Example Unit 1 2 3*) 4*) 5*) 6*) 7*) 8*) 9 10 11 12 Isocyanate % 100 95.000 95.000 95.000 95.000 97.787 95.000 95.000 95.000 95.00 95.00 95.00 95.00 prepolymer 1 Basolite A 100 % 5.000 Basolite A 120 % 5.000 Aerosil R 504 % 5.000 Omyacarb 2T- % 5.000 AV 2-Ethyl-1- % 2.213 hexanol Purolite A110 % 5.000 Lewatit VP OC % 5.000 1065 Aerosil VPR % 5.000 511 Mg-DHT % 5.00 Mg-HT % 5.00 Al-DHT % 5.00 Al-HT % 5.00 NCO content % 6.09 5.52 4.78 5.52 5.69 5.39 5.86 5.70 4.47 5.43 5.59 5.32 5.46 Total monomer % 2.5 1.6 0.05 1.5 1.9 1.13 1.38 1.51 0.72 1.1 1.8 1.5 1.6 content 2.4-TDI % 0.4 0.1 0.02 0.1 0.2 0.03 0.08 0.11 0.02 0.1 0.2 0.1 0.1 content 2.6-TDI % 2.1 1.5 0.03 1.4 1.7 1.1 1.3 1.4 0.7 1.0 1.6 1.4 1.5 content Monomer rel % −36 −98 −40 −24 −55 −45 −40 −71 −56 −28 −40 −36 reduction Viscosity Pas 11.4 22.7 21.6 82.4 15.6 17.1 13.1 13.7 54.5 15.6 16.4 18.2 16.8 (25° C.; 10 s.sup.−1) Viscosity Pas 11.3 21.4 20.9 59.1 15.1 16.8 12.8 13.4 44.1 15.2 16.1 17.8 16.5 (25° C.; 40 s.sup.−1) *)comparative example

[0116] The same absorbents were tested in examples 13-24 with the MDI based isocyanate prepolymer 2 having a NCO value of 5.27% and a residual MDI monomer content of 7.2%. The NCO content could be reduced to 4.16% and residual MDI monomer content of 2.5% in example 14 which contains Basolite A 120; the performance of the other absorbents in example 13 and 15-24 was worse. These examples demonstrate that Basolite A 120 in example 14 has again the highest efficiency to capture MDI monomers which is in analogy to the TDI prepolymer examples 1-12.

TABLE-US-00003 TABLE 2 Examples 13-24 using isocyanate prepolymer 2 based on MDI. Example Unit 13 14 15*) 16*) 17*) 18*) 19*) 20*) 21 22 23 24 Isocyanate % 100 95.000 95.000 95.000 95.000 96.964 95.000 95.000 95.000 95.00 95.00 95.00 95.00 prepolymer 2 Basolite A % 5.000 100 Basolite A % 5.000 120 Aerosil R 504 % 5.000 Omyacarb % 5.000 2T-AV 2-Ethyl-1- % 3.035 hexanol Purolite % 5.000 A110 Lewatit VP % 5.000 OC 1065 Aerosil VPR % 5.000 511 Mg-DHT % 5.00 Mg-HT % 5.00 Al-DHT % 5.00 Al-HT % 5.00 NCO content % 5.27 4.97 4.16 4.72 5.25 4.20 4.59 4.96 4.37 4.13 4.51 4.52 4.54 Total % 7.2 5.2 2.5 5.5 6.2 3.7 4.7 5.5 4.5 4.2 6.0 5.3 5.3 monomer content 2,2′-MDI % 0.4 0.4 0.3 0.4 0.4 0.3 0.4 0.4 0.4 0.5 0.4 0.4 0.4 content 2,4′-MDI % 4.7 3.5 2.1 3.7 4.1 2.6 3.2 3.7 3.1 2.8 4.0 3.6 3.6 content 4,4′-MDI % 2.1 1.3 0.1 1.4 1.7 0.8 1.1 1.4 1.0 0.9 1.6 1.3 1.3 content Monomer rel % −28 −65 −24 −14 −49 −35 −24 −38 −42 −17

−26 −26 reduction Viscosity Pas 2.8 6.5 6.9 12.3 4.5 5.3 4.0 4.0 9.9 4.6 5.1 5.1 4.9 (60° C.; 10 s.sup.−1) Viscosity Pas 2.8 5.9 6.1 12.0 4.2 5.1 3.9 3.9 9.4 4.4 4.8 5.0 4.7 (60° C.; 40 s.sup.−1) *)comparative example

[0117] Additionally to the MOF type absorbents the monomer removal using zeolites was also tested with isocyanate prepolymer 1 and isocyanate prepolymer 2.

Description of the Raw Materials

[0118] Zeolite MU 0169: Deboronated beta, 42% Si/0.05% B, BET 488 m.sup.2/g

[0119] Zeolite DY 43: HY, 718 m2/g

[0120] Zeolite CBV 100: NaY, SiO.sub.2:Al.sub.2O.sub.3=5, 727 m.sup.2/g

[0121] Zeolite CBV 760: HY, SiO.sub.2:Al.sub.2O.sub.3=60, 720 g/m.sup.2

[0122] All zeolites were dried under vacuum at 80° C. overnight before use. After drying the absorbents were directly used for the prepolymer/absorbent mixtures. Typical preparation procedure: 47.5 g of isocyanate prepolymer 1 or 2 was charged into a PE cup, closed with a lid and heated to 60° C. in a laboratory heating oven. 2.5 g of dried absorbent was added and homogenized with a Vollrath laboratory mixer. The samples were filled into airtight glass vials and annealed as follows. Example 25 and 26 were annealed after 1 day for 3 hours at 60° C. and then stored for 3 days at room temperature before analysis. Example 27 and 28 were stored at 60° C. for 1 hour, then kept at room temperature for 6 days and additionally annealed for 2 hours at 60° C., those with isocyanate prepolymer 2 at 60° C. for 2 hours. Example 29 and 30 were stored at room temperature for 14 days. Example 31 and 32 were stored at 60° C. for 1 hour, then kept at room temperature for 6 days and additionally annealed for 2 hours at 60° C. NCO content was determined by titration, viscosity measured with a Brookfield viscosimeter with cone-plate geometry and monomer content was determined with a HPLC method as described earlier.

[0123] The results in Table 3 and 4 clearly demonstrate that the ability of the zeolites to capture the free isocyanate monomer from prepolymers is relatively low compared to the Basolite A 120 in example 2 and 14.

[0124] For some cases, e.g. example 27, 28, 31 and 32, the monomer content was not even determined since the measured NCO content of the mixture was so close to the theoretical NCO of the plain mixture. As the Basolite references show the drop of NCO content is an essential indicator for the removal of the monomeric isocyanate.

[0125] Furthermore, in Example 29 with Basolite A 120 and the MDI prepolymer 1 the monomer removal even takes place also at room temperature granting 14 days of storage time. No such results were obtained with zeolites even not at thermal annealing at 60° C. for 3 hours.

TABLE-US-00004 TABLE 3 Examples 25-28 using zeolites and isocyanate prepolymer 1 based on TDI. Example Unit 25 26*) 27*) 28*) Isocyanate % 100 95.0 95.0 100 95.0 95.0 prepolymer 1 Basolite % 5.0 0.000 A 120 Zeolite % 0.000 5.0 MU 0169 Zeolite % 0.000 0.000 5.0 DY 43 Zeolite % 0.000 0.000 5.0 CBV 100 NCO content % 6.22 4.60 5.89 6.24 5.70 5.77 Theor. NCO — 5.91 5.91 — 5.93 5.93 content for mixture Total % 2.8 <0.1 2.2 2.5 n/a n/a monomer content 2,4-TDI % 0.5 <0.1 0.3 0.3 n/a n/a content 2,6-TDI % 2.3 <0.1 1.9 2.2 n/a n/a content Monomer rel % — −96 −21 — n/a n/a reduction Viscosity Pas 11.4 21.8 15.2 10.8 14.4 16.2 (25° C.), Brookfield

*)Comparative example

TABLE-US-00005 TABLE 4 Examples 29-32 using zeolites isocyanate prepolymer 2 based on MDI. Example Unit 29 30*) 31*) 32*) Isocyanate % 100 95.0 95.0 95.0 95.0 prepolymer 2 Basolite A 120 % 5.0 Zeolite CBV 760 % 5.0 Zeolite DY 43 % 5.0 Zeolite CBV 100 % 5.0 NCO content % 4.60 3.12 4.18 4.08 4.29 Theor. NCO content — 4.37 4.37 4.37 4.37 for mixture Total monomer % 5.7 0.8 4.4 n/a n/a content 2,4′-MDI + 2,2′-MDI % 0.3 0.7 3.5 n/a n/a content 4,4′-MDI content % 2.2 0.1 0.9 n/a n/a Monomer reduction rel % — −86 −23 n/a n/a Viscosity (60° C.), Pas 3.9 6.8 4.9 7.1 7.0 Brookfield *)comparative example

4. Application Tests in PU Hot Cast Elastomers

[0126] The demonomerized isocyanate prepolymer 1 was prepared following the above described process in two mixing ratios: 95 wt % isocyanate prepolymer 1 with 5 wt % Basolite A120 and 97 wt % isocyanate prepolymer 1 with 3 wt % Basolite A 120. All three prepolymers including the untreated isocyanate prepolymer 1 were heated to 80° C. in a laboratory oven, the curing agent 3,5-Dimethylthio-toluenediamine was used at room temperature. Both components were charged in the given weight ratio into a PE container and homogenized using a SpeedMixer™ under vacuum for 20 sec at 1820 rpm. The reaction mixture was then poured into a preheated aluminum mold at 90° C. and post-cured for 24 hours at 90° C. Mechanical testing included Shore A and D hardness (DIN ISO 7619-1), tensile strength and elongation at break with S2 test specimen (DIN 53504), tear strength with Graves geometry (DIN ISO 34-1, B), E modulus with S2 test specimen, compression set with 10% compression (DIN ISO 815-1) and abrasion (DIN ISO 4649).

[0127] The mechanical properties of the obtained hot cast elastomers are shown in Table 5. The mechanical performance of the elastomers with isocyanate prepolymer 1 with <0.1% TDI (example 34) or 0.3% TDI (example 35) remains below the reference (example 33) since the monomer was captured by the Basolite A 120 scavenger. The absence of isocyanate monomers leads to the situation that no hard segments can be formed which typically are strongly influencing the final mechanical properties in polyurethane elastomers. However, this observation is well known for monomer-free prepolymers used for polyurethane elastomers independent of the method of demonomerization. The examples 34-35 illustrate that prepolymers which were demonomerized with Basolite A 120 can be processed and used for hot cast elastomer applications.

TABLE-US-00006 TABLE 5 Polyurethane hot cast elastomers. Example 25 26 27 Unit 33 34 35 3,5-Dimethylthio- g 41.79 32.83 36.01 toluenediamine (CAS 106264-79-3) Isocyanate prepolymer 1 g 278.21 0 0 Isocyanate prepolymer 1 + g 0 287.17 0 5 wt % Basolite A120 Isocyanate prepolymer 1 + g 0 0 284.0 3 wt % Basolite A120 NCO % wt % 6.09 4.59 5.09 Monomer content [wt %] wt % 2.5 0.07 0.3 Mechanical properties Shore D 65 69 40 Tensile strength MPa 68 28 35 Elongation % 490 370 450 Tear strength (Graves) kN/m 73 50 76 E-Modulus MPa 62 59 51 Compression set % 27 22 28 (72 h/23° C./30 min) Compression set % 59 53 56 (24 h/70° C./30 min) Abrasion mm.sup.3 53 67 64

Claims

1. A process for the reduction of the content of residual monomeric isocyanate in a polyurethane prepolymer comprising said residual monomeric isocyanate comprising the step of (i) contacting the polyurethane prepolymer with a porous material having pores formed by a framework comprising isocyanate reactive functional groups under conditions allowing chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores to covalently bind the reacted isocyanate, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.

2. The process according to claim 1, wherein the residual monomeric isocyanate is an aromatic or aliphatic isocyanate.

3. The process according to claim 1, wherein before contacting the polyurethane prepolymer with the porous material, the porous material has a mean pore size in the range of from 0.35 to 2.1 nm according to DIN ISO 9277:2014-01.

4. The process according to claim 1, wherein before contacting the polyurethane prepolymer with the porous material, the porous material has a pore volume in the range of from 0.04 to 1.7 cm.sup.3/g according to DIN ISO 9277:2014-01.

5. The process according to claim 1, wherein before contacting the polyurethane prepolymer with the porous material, the porous material has a specific surface area BET in the range of from 10 to 5700 m.sup.2/g according to DIN ISO 9277:2014-01.

6. The process according to claim 1, wherein the isocyanate reactive functional groups are selected

from the group consisting of amino, hydroxyl, thiol, epoxy and a mixture of two or more of these groups.
7. The process according to claim 1, wherein the polyurethane prepolymer is liquid under the conditions allowing the chemical reaction.
8. The process according to claim 1, wherein the ratio (w: w) of polyurethane prepolymer to porous material is in the range of from 100:0.1 to 100:50.
9. The process according to claim 1, wherein the residual monomer isocyanate content of the polyurethane prepolymer is reduced by at least 50%.
10. The process according to claim 9, wherein the porous metal-organic framework material comprising the at least one metal ion and the at least one at least bidentate organic compound, which is coordinately bound to said metal ion is prepared by contacting the at least one metal ion with the at least one at least bidentate organic compound and wherein the at least one at least bidentate organic compound comprises the isocyanate reactive functional groups.
11. The process of according to claim 9, wherein the at least one metal ion is a metal ion of a metal selected from the group consisting of Mg, Ca, Sr, Ba, Sc, Y, Ln, Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W, Mn, Re, Fe, Ro, Os, Co, Rh, Ir, Ni, Pd, Pt, Cu, Ag, Au, Zn, Cd, Hg, Al, Ga, In, TI, Si, Ge, Sn, Pb, As, Sb, Bi, and mixtures thereof, wherein Ln represents a lanthanide.
12. The process according to claim 1, wherein the at least one at least bidentate organic compound is derived from an amino polycarboxylic acid.
13. A polyurethane prepolymer composition, wherein the composition comprises a polyurethane prepolymer comprising a porous material having pores formed by a framework comprising isocyanate reactive functional groups, wherein the reactive functional groups are reacted with monomeric isocyanate in at least a fraction of the pores, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.
14. A method of using a porous material having pores formed by a framework, comprising isocyanate reactive functional groups, the method comprising using the porous material to reduce the content of residual monomeric isocyanate in a polyurethane prepolymer by chemical reaction of the reactive functional groups with the residual monomeric isocyanate in at least a fraction of the pores, wherein the porous material is a porous metal-organic framework material comprising at least one metal ion and at least one at least bidentate organic compound, which is coordinately bound to said metal ion.
15. The process according to claim 1, wherein the isocyanate reactive functional groups are selected from the group consisting of amino, hydroxyl, thiol, epoxy and a mixture of two or more of amino groups.
16. The process according to claim 1, wherein the ratio (w: w) of polyurethane prepolymer to porous material is in the range of from 100:0.1 to 100:20.
17. The process according to claim 1, wherein the ratio (w: w) of polyurethane prepolymer to porous material is in the range of from 100:0.1 to 100:10.
18. The process according to claim 1, wherein the residual monomer isocyanate content of the polyurethane prepolymer is reduced by at least 90%.
19. The process according to claim 1, wherein the residual monomer isocyanate content of the polyurethane prepolymer is reduced by at least 95%.
20. The process according to claim 1, wherein the at least one at least bidentate organic compound is derived from an amino dicarboxylic acid.